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Modular antenna with mounting module

Abstract

A mounting apparatus for a modular antenna is described, wherein the mounting apparatus is attachable to a plurality of different radiation apparatus or modules and/or to a plurality of different feeding apparatus or modules. The mounting apparatus has a surface facing a measurement direction, wherein the surface is configured such that a surface parameter of the mounting apparatus is smaller than a predefined threshold, wherein the surface parameter is associated with a ratio of a first projected area and a second projected area. The first projected area corresponds to an area confined by the surface of the mounting apparatus facing the measurement direction and projected onto a plane being perpendicular to the measurement direction. The second projected area corresponds to an aperture of the modular antenna projected onto the plane being perpendicular to the measurement direction. Further, a modular antenna and an over-the-air measurement system are described.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) Embodiments of the present disclosure generally relate to a mounting module for a modular antenna. Further, embodiments of the present disclosure relate to a modular antenna as well as to an over the air (OTA) measurement system.

BACKGROUND

(2) In the state of the art, modular antennas and OTA measurement systems are known. Typically, measurement applications require various antenna radiation characteristics. However, there is no antenna that is optimal for all applications, as the required radiation characteristics can be contradictory at times. Antenna characteristics are mainly defined by the radiation part of the antenna. Hence, most OTA measurement systems require multiple different types of antennas. However, many basic measurement antennas comprise two parts, namely a feeding part and a radiation part. These two parts are often integrated without the possibility of detachment.

(3) In order to reduce costs, modular antennas are known. These antennas comprise an interface flange which provides for mounting modularity and reduces the amount of parts needed for an OTA measurement system. However, the mounting part of modular antennas has a substantial surface area in relation to the antenna cross section. Hence, the cross section of the mounting module can lead to unwanted reflections towards a device under test (DUT), e.g., an antenna under test (AUT), as well as to the creation of a standing wave between the DUT and the measurement antenna. The resulting reflections and standing wave effect decrease the measurement accuracy and influence the results of the OTA measurement system.

(4) Therefore, there is a need for a modular antenna which can be used in OTA measurement systems while providing the same amount of accuracy as a common antenna would. Further, there

is a need to decrease the costs of OTA measurement systems.

SUMMARY

(5) Embodiments of the present disclosure provide a mounting module for a modular antenna with the mounting module being attachable to a plurality of different radiation modules and/or to a plurality of different feeding modules. In an embodiment, the mounting module has a surface facing towards a measurement direction, wherein the surface of the mounting module facing towards the measurement direction is configured such that a surface parameter of the mounting module is smaller than a predefined threshold. The surface parameter is associated with a ratio of a first projected area and a second projected area. The first projected area corresponds to an area confined by the surface of the mounting module facing towards the measurement direction and projected onto a plane which is perpendicular to the measurement direction. The second projected area corresponds to an aperture of the modular antenna which is projected onto the plane perpendicular to the measurement direction.

(6) The present disclosure is based on the main idea that an electromagnetic cross section, for example a radar cross section, of a mounting module is limited by reducing the transverse surfaces (of components involved) towards a measurement direction in relation to the antenna aperture.

(7) In some embodiments, the first projected area is a measure for the portion of the surface of the mounting module extending transversally to the measurement direction, i.e., for the portion of the surface that mainly causes perturbations during measurements.

(8) Accordingly, the surface of the mounting module may be configured such that the surface parameter is reduced compared to common mounting modules, for example such that the surface parameter is as small as possible.

(9) Accordingly, an absolute reduction is obtained, namely an absolute reduction of the transverse surfaces of the mounting module, namely its surface area. Consequently, unwanted reflections towards a device under test (DUT), e.g. an antenna under test (AUT), as well as to the creation of a standing wave between the DUT and the measurement antenna can be avoided.

(10) Embodiments of the present disclosure further provide a modular antenna comprising a mounting module according to the present disclosure, a feeding module, and a radiation module.

(11) Furthermore, embodiments of the present disclosure provide an OTA measurement system comprising at least one modular antenna according to the present disclosure.

(12) Examples of the mounting module according to the present disclosure provide increased system flexibility in regards to modular antennas and OTA measurement systems. Since a variety of combinations between feeding modules and radiation modules is possible due to the mounting module, the costs of the overall system are reduced as well.

(13) In some embodiments, the radiation module of the modular antenna can be changed according to the required radiation characteristics provided by the present OTA measurement without having to replace the feeding module of the modular antenna.

(14) Moreover, the measurement accuracy and results of the OTA measurement system are improved, since the reduced electromagnetic cross section of the disclosed mounting module prevents the creation of standing waves, which could possibly impact the measurement results negatively. Further, reflections impairing the OTA measurements are reduced.

(15) In order to improve the results, the predefined threshold of the surface parameter in some embodiments is smaller than or equal to 3.5. Even more accurate measurement results can be obtained, when the predefined threshold of the surface parameter is smaller than or equal to 3. In some embodiments, the predefined threshold of the surface parameter is a number smaller than or equal to 1, since this ensures the most accurate measurement results of the OTA measurement system.

(16) The possibility of standing waves is decreased even further by covering all portions of the mounting module that extend transverse to the measurement direction with an absorber. While there are many different ways how the absorber can be arranged, using absorbers having triangular

shapes has proved most effective.
(17) Besides the standing wave effects, interference between the signal used for measurements and reflected electromagnetic waves has also been shown to decrease the measurement accuracy. The reflection coefficient of dielectric materials with a relative permittivity smaller than 1.5 was found to be negligible. For that reason, the mounting module may at least partially comprise of a dielectric material having a dielectric constant smaller than or equal to, for example, 1.5.
(18) In some embodiments, only a portion of the mounting module having a dielectric constant greater than 1.5 may be considered for the surface parameter. Accordingly, portions of the mounting module consisting of the dielectric material having a dielectric constant smaller than or equal to 1.5 do not contribute to the surface parameter.
(19) Further, portions of the mounting module, which could lead to reflections of electromagnetic waves to other parts of the mounting module, are also covered with an absorber. Accordingly, multiple reflections of electromagnetic waves off the mounting module are reduced such that the accuracy of measurement results is further improved.
(20) In order to provide a secure connection between the feeding module and the radiation module, the mounting module may comprise at least one flange. Hence, the mounting module is attachable to a plurality of different radiation modules and/or to a plurality of different feeding modules by the at least that one flange.
(21) The costs resulting from the need of different antennas to be used in an OTA measurement system can be avoided by the mounting module being standardized. That way, it can be ensured that all feeding modules fit all radiation modules with the help of the mounting module.
(22) According to one embodiment of the present disclosure, the mounting module can be integrated into the feeding module or into the radiation module. This still provides a flexibility regarding the antenna types while reducing the number of parts needed.
(23) An alternative embodiment of the present disclosure provides the mounting module as a part, which is established separately from the feeding module and the radiation module. This ensures that all combinations between different radiation and feeding modules are possible, without having to worry whether a feeding module with an integrated mounting module has to be combined with a radiation module having an integrated mounting module.
(24) In some embodiments, a surface area and/or a cross section of the feeding module may be circular or rectangular. Since those are the most common shapes of feeding modules for antennas, they do not limit the flexibility provided by the mounting module.
(25) Further, the mounting module is connected with the radiation module and/or with the feeding module through a thread, a bayonet mount, at least one screw, at least one bolt, or a snap-in in order to provide a modular antenna.
(26) Moreover, other types of (detachable) connections may also be used for connecting the mounting module with the radiation module and/or with the feeding module, e.g., a compression fitting, a flare fitting, a flange fitting, a push fitting, a rubber sleeve fitting or a bushing connector. However, it is to be understood that any other suitable type of connection may be used.

Description

DESCRIPTION OF THE DRAWINGS
(1) The foregoing aspects and many of the attendant advantages of the claimed subject matter will become more readily appreciated as the same become better understood by reference to the following detailed description, when taken in conjunction with the accompanying drawings, wherein:
(2) FIG. **1** schematically shows an example of an OTA measurement system according to an embodiment of the present disclosure, and comprises a modular antenna according to the present

disclosure;
(3) FIG. **2** schematically illustrate a side view of a modular antenna according to an embodiment of the present disclosure;
(4) FIGS. **3**A to **3**E schematically illustrate side views of various modular antennas according to one or more embodiments of the present disclosure, respectively;
(5) FIGS. **4**A to **4**I schematically illustrate examples of a radiation module of a modular antenna according to an embodiment of the present disclosure, respectively;
(6) FIGS. **5**A and **5**B schematically illustrate a side view and a top view of a state of the art modular antenna, respectively;
(7) FIGS. **6**A and **6**B schematically illustrate a side view and a top view of a modular antenna according to an embodiment of the present disclosure, respectively; and
(8) FIGS. **7**A to **7**D schematically illustrate a variety of fasteners, which can be used to connect the different modules of the modular antenna according to embodiments of the present disclosure.
DETAILED DESCRIPTION
(9) The detailed description set forth below in connection with the appended drawings, where like numerals reference like elements, is intended as a description of various embodiments of the disclosed subject matter and is not intended to represent the only embodiments. Each embodiment described in this disclosure is provided merely as an example or illustration and should not be construed as preferred or advantageous over other embodiments. The illustrative examples provided herein are not intended to be exhaustive or to limit the claimed subject matter to the precise forms disclosed.
(10) FIG. **1** depicts an over the air (OTA) measurement system **10** that is used for testing an antenna under test (AUT) **12**. Generally, the over the air (OTA) measurement system **10** may be used to test a device under test that is different to an antenna under test **12**, e.g. a device having an antenna.
(11) In the embodiment shown, the OTA measurement system **10** comprises a shielded chamber **14** that accommodates the AUT **12** as well as a modular antenna **16** that is used for testing the AUT **12**. The modular antenna **16** is part of the OTA measurement system **10**. In general, the shielded chamber **14** is configured to shield an interior space of the shielded chamber **14** from external electromagnetic waves that may disturb or rather affect the measurements, thereby lowering the accuracy of the measurement results.
(12) In some embodiments, the shielded chamber **14** may be an anechoic chamber, i.e., inner surfaces of the shielded chamber **14** may be covered with anti-reflection components, such as absorbers, that are configured to reduce reflections within the interior space of the shielded chamber.
(13) The OTA measurement system **10** may also comprise a positioner system **18** for aligning the AUT **12** as well as two reflectors **20**.
(14) The OTA system **10** shown in the embodiment of FIG. **1** is a multi-reflector system, but any other type of OTA measurement system can be used as well.
(15) One example of the modular antenna **16** used in the OTA measurement system **10** is shown in FIG. **2**. As shown in FIG. **2**, the modular antenna **16** comprises a feeding module **22**, a radiation module **24**, and a mounting module **26**. In the example embodiment shown in FIG. **2**, the mounting module **26** is integrated into the feeding module **22**, while the radiation module **24** is secured to the mounting module **26** through two bolts **28**. Instead of bolts **28**, bars, rods or similar may be used.
(16) Depending on the measurement application, different kinds of radiation characteristics are needed. These are mainly defined by the radiation module **24** of the modular antenna **16**. Some of the characteristics that are needed for OTA measurement systems are broadband antennas, antennas with low, medium, or high directivity and antennas with a high radial cross-section and low side-lobes, as shown in FIG. **3**A to **3**E, respectively.
(17) In general, a surface area of the feeding module **22** may be circular or rectangular. Likewise, a

cross section of the feeding module **22** may be circular or rectangular. Other shapes may be practiced with embodiments of the present disclosure.
(18) The radiation module **24** depicted in FIG. **2** is only an example. Any other type of radiation module **24** like a wire dipole, a loop, a wire monopole, a Yagi-Uda ray, a horn, a microstrip patch, a corner reflector, a slot, or a parabolic reflector (dish), as shown in FIGS. **4**A to **4**I respectively, can be used as well.
(19) Further, the connection between the different modules can also be given by means of threads **40**, bayonet mounts **42**, snap-ins **44**, or screws **46**, as is shown in FIGS. **7**A to **7**D.
(20) Other types of (detachable) connections be used for connecting the mounting module **26** with the radiation module **24** and/or with the feeding module **22** comprise a compression fitting, a flare fitting, a flange fitting, a push fitting, a rubber sleeve fitting, a bushing connector, etc.
(21) Generally, the mounting module **26** is capable of being attached to a plurality of different radiation modules and/or to a plurality of different feeding modules, thereby ensuring the modularity for the modular antenna **16**.
(22) In some embodiments, the mounting module **26** of the modular antenna **16** can also at least partially be covered with an absorber **30**, as is shown e.g., in FIG. **6**A.
(23) In order to improve the accuracy of measurement results obtained by OTA measurement systems, e.g., by the OTA measurement system **10** depicted in FIG. **1**, the amount of standing waves and unwanted reflected signals has to be reduced.
(24) As is illustrated in FIGS. **6**A, and **6**B, this is achieved by limiting an electromagnetic cross section of the mounting module **26** compared to common mounting modules known in the state of the art of FIGS. **5**A, **5**B. It will be appreciated that the radiation module **24** is not shown in FIGS. **5**A, **5**B, **6**A, and **6**B for illustration purposes.
(25) This is done by reducing transverse surfaces of the mounting module **26** facing towards a measurement direction M in relation to an antenna aperture **34**. For example, a surface parameter R_fa is reduced in order to be smaller than a predefined threshold, e.g., smaller than or equal to 3.5, smaller than or equal to 3 or even smaller than or equal to 1.
(26) In some embodiments, the surface parameter R_fa is minimized.
(27) The surface parameter R_fa is associated with a ratio of a first projected area and a second projected area, and is given by the following equation:
(28) R_fa = (∫∫n_f .Math. n_mdA_f) / (∫∫n_a .Math. n_mdA_a) where n_m is a unit vector in measurement direction M, n_a is a unit vector in a direction normal to the antenna aperture **34**, and n_f is a unit vector in a direction normal to the surface of the mounting module **26**.
(29) Therein the integral ∫∫custom charactercustom charactern_f.Math.n_m dA_f custom character is evaluated over an area A_f confined by the surface of the mounting module **26** facing towards the measurement direction M.
(30) The integral ∫∫custom charactercustom charactern_a.Math.n_m dA_acustom character is evaluated over the aperture **34** having an area A_a.
(31) Accordingly, the first projected area, namely the numerator of above-mentioned fraction defining the surface parameter R_fa, corresponds to the area confined by the surface of the mounting module **26** facing towards the measurement direction M and projected onto a plane that is perpendicular to the measurement direction M.
(32) Further, the second projected area, namely the denominator of above-mentioned fraction defining the surface parameter R_fa, corresponds to the aperture **34** of the modular antenna **16** projected onto a plane being perpendicular to the measurement direction M.
(33) The first projected area and the second projected area are put in relation with each other, thereby defining the ratio.
(34) This ratio can, for example, be effectively minimized by placing the surface of the mounting module **26** at an angle θ. Steep angles θ in regards to measurement direction M are preferred, since they have the biggest impact when it comes to reducing the surface parameter R_fa.

(35) While surface parameter R_fa relates to the placement of the surface area A_a of the antenna aperture **34** and surface area A_f of the mounting module **26**, during operation, a reduced surface parameter R_fa leads to an increased signal strength. By having slanted mounting surfaces a flux of unwanted reflected electromagnetic waves is reduced, which could interfere with the measurement signal.
(36) However, sometimes it is unavoidable to have a transverse mounting surface **36** towards the measurement direction M. In that case, it is advisable to use dielectric materials with a relative permittivity smaller than 1.5. These materials are transparent enough in regards to electromagnetic radiation as to not reflect electromagnetic waves, i.e., they are transparent in regards to electromagnetic waves and therefore let them pass through. For that particular reason, dielectric materials having a dielectric constant smaller than or equal to 1.5 are assumed to not contribute to the surface parameter R_fa.
(37) Alternatively and/or additionally, the mounting surface **36** can at least partially be covered by absorbers **30**. It has been found that the most effective absorber shape for electromagnetic waves is a pyramid module for low frequencies while a cone shape is nearly as effective at higher frequencies.
(38) Generally, absorbers can be considered as a vertically gradient structure. In order to preserve the flexibility of the OTA measurement system **10**, the absorbers **30** may be broadband absorbers. Generally, though, it is also possible to use resonant absorbers.
(39) Since multiple modular antennas **16** might be used in the OTA measurement system **10**, all portions of the mounting module **26** which might lead to reflections of electromagnetic waves should be covered with absorbers **30**. In other words, also surfaces that might not directly contribute to the surface parameter R_fa should be taken into consideration when placing absorbers **30**.
(40) Further, mounting module **26** can be standardized, such that it is universally usable for different feeding modules **22** and radiation modules **24** as already indicated above. This is especially cost-effective, since the feeding module **22** is usually the most complicated and expensive part of the modular antenna **16**. By having a standardized mounting module **26**, it would be possible to only replace the feeding module **22** when it is absolutely necessary and not each time a different antenna type is needed. Instead, it would be possible to simply switch out the radiation module **24** of the modular antenna **16** which is usually the cheaper solution.
(41) The present application may reference quantities and numbers. Unless specifically stated, such quantities and numbers are not to be considered restrictive, but exemplary of the possible quantities or numbers associated with the present application. Also in this regard, the present application may use the term “plurality” to reference a quantity or number. In this regard, the term “plurality” is meant to be any number that is more than one, for example, two, three, four, five, etc. The terms “about,” “approximately,” “near,” etc., mean plus or minus 5% of the stated value. For the purposes of the present disclosure, the phrase “at least one of A and B” is equivalent to “A and/or B” or vice versa, namely “A” alone, “B” alone or “A and B.”. Similarly, the phrase “at least one of A, B, and C,” for example, means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B, and C), including all further possible permutations when greater than three elements are listed.
(42) The principles, representative embodiments, and modes of operation of the present disclosure have been described in the foregoing description. However, aspects of the present disclosure which are intended to be protected are not to be construed as limited to the particular embodiments disclosed. Further, the embodiments described herein are to be regarded as illustrative rather than restrictive. It will be appreciated that variations and changes may be made by others, and equivalents employed, without departing from the spirit of the present disclosure. Accordingly, it is expressly intended that all such variations, changes, and equivalents fall within the spirit and scope of the present disclosure, as claimed.

Claims

1. A mounting apparatus for a modular antenna, the mounting apparatus being attachable to a plurality of different radiation modules and/or to a plurality of different feeding modules, wherein the mounting apparatus, a radiation module of the plurality of different radiation modules, and a feeding module of the plurality of different feeding modules are combinable to form the modular antenna, wherein the radiation module defines radiation characteristics of the modular antenna, wherein the mounting apparatus has a surface facing towards a measurement direction, wherein the surface of the mounting apparatus facing towards the measurement direction is configured such that a surface parameter of the mounting apparatus is smaller than a predefined threshold, wherein the surface parameter is associated with a ratio of a first projected area and a second projected area, wherein the first projected area corresponds to an area confined by the surface of the mounting apparatus facing towards the measurement direction and projected onto a plane that is perpendicular to the measurement direction, wherein the second projected area corresponds to an aperture of the modular antenna projected onto the plane being perpendicular to the measurement direction, and wherein the predefined threshold is a number being smaller than or equal to 3.5.
2. The mounting apparatus of claim 1, wherein the predefined threshold is a number being smaller than or equal to 3.
3. The mounting apparatus of claim 1, wherein the predefined threshold is a number being smaller than or equal to 1.
4. The mounting apparatus according to claim 1, wherein portions of the mounting apparatus that extend transverse to the measurement direction are covered with an absorber.
5. The mounting apparatus according to claim 1, wherein the mounting apparatus at least partially comprises a dielectric material having a dielectric constant smaller than or equal to 1.5.
6. The mounting apparatus of claim 1, wherein portions of the mounting apparatus leading to reflections of electromagnetic waves to other portions of the mounting apparatus are covered with an absorber.
7. The mounting apparatus according to claim 1, wherein the mounting apparatus comprises at least one flange, wherein the mounting apparatus is attachable to the plurality of different radiation modules and/or to the plurality of different feeding modules by the at least one flange.
8. The mounting apparatus according to claim 1, wherein the mounting apparatus is standardized.
9. A modular antenna comprising: a mounting apparatus; a feeding module selected from a plurality of different feeding modules; and a radiation module selected from a plurality of different radiation modules; wherein the mounting apparatus, the radiation module, and the feeding module constitute the modular antenna, wherein the radiation module defines radiation characteristics of the modular antenna, wherein the mounting apparatus is attachable to the plurality of different radiation modules and/or to the plurality of different feeding modules, wherein the mounting apparatus has a surface facing towards a measurement direction, wherein the surface of the mounting apparatus facing towards the measurement direction is configured such that a surface parameter of the mounting apparatus is smaller than a predefined threshold, wherein the surface parameter is associated with a ratio of a first projected area and a second projected area, wherein the first projected area corresponds to an area confined by the surface of the mounting apparatus facing towards the measurement direction and projected onto a plane that is perpendicular to the measurement direction, wherein the second projected area corresponds to an aperture of the modular antenna projected onto the plane being perpendicular to the measurement direction, and wherein the predefined threshold is a number being smaller than or equal to 3.5.
10. The modular antenna according to claim 9, wherein the mounting apparatus is integrated into the feeding module or into the radiation module.
11. The modular antenna according to claim 9, wherein the mounting apparatus is established

separately from the feeding module and the radiation module.
12. The modular antenna according to claim 9, wherein a surface area and/or a cross section of the feeding module is circular or rectangular.
13. The modular antenna according to claim 9, wherein the mounting apparatus is connected with the radiation module and/or with the feeding module through a thread, a bayonet mount, at least one screw, at least one bolt, or a snap-in.
14. An over-the-air (OTA) measurement system comprising a modular antenna, the modular antenna comprising: a mounting apparatus; a feeding module selected from a plurality of different feeding modules; and a radiation module selected from a plurality of different radiation modules; wherein the mounting apparatus, the radiation module, and the feeding module constitute the modular antenna, wherein the radiation module defines radiation characteristics of the modular antenna, wherein the mounting apparatus is attachable to the plurality of different radiation modules and/or to the plurality of different feeding modules, wherein the mounting apparatus has a surface facing towards a measurement direction, wherein the surface of the mounting apparatus facing towards the measurement direction is configured such that a surface parameter of the mounting apparatus is smaller than a predefined threshold, wherein the surface parameter is associated with a ratio of a first projected area and a second projected area, wherein the first projected area corresponds to an area confined by the surface of the mounting apparatus facing towards the measurement direction and projected onto a plane that is perpendicular to the measurement direction, wherein the second projected area corresponds to an aperture of the modular antenna projected onto the plane being perpendicular to the measurement direction, and wherein the predefined threshold is a number being smaller than or equal to 3.5.